Soumiya Ayyappan_W6_D3

Code and its output:

Step 1:

Code:

```
day1 > codecod2.py > ...
  1  from pyspark.sql import SparkSession
  2  from pyspark.sql.functions import *
  3  from pyspark.sql.types import *
  4
  5  spark=SparkSession.builder.appName("EmployeeData").getOrCreate()
  6  emp_path="day1/data/employee.csv"
  7  per_path="day1/data/performance.csv"
  8  employees_df=spark.read.csv(emp_path,header=True,inferSchema=True)
  9  performance_df=spark.read.csv(per_path,header=True,inferSchema=True)
 10  employees_df=employees_df.withColumn("employee_id",col("employee_id").cast(IntegerType()))
 11  performance_df=performance_df.withColumn("employee_id",col("employee_id").cast(IntegerType()))
 12  employees_df.show()
 13  performance_df.show()
```

Output:

```
(.venv) default:~/pyspark_training$ /home/kasm-user/pyspark_training/.venv/bin/python /home/kasm-user/pyspark_training/day1/codecod2.py
e
+-----------+----------+---------+-------------+
|employee_id|first_name|last_name|    full_name|
+-----------+----------+---------+-------------+
|        101|     Aarav|   Sharma| Aarav Sharma|
|        102|     Priya|    Patel|         NULL|
|        103|     Rohan|    Mehta|  Rohan Mehta|
|        104|    Ananya|    Verma|         NULL|
|        105|    Vikram|    Joshi| Vikram Joshi|
|        106|      Neha|   Kapoor|         NULL|
|        107|     Arjun| Malhotra|Arjun Malhotra|
|        108|     Sneha|     Iyer|         NULL|
|        109|     Karan|    Desai|  Karan Desai|
|        110|     Meera|     Nair|         NULL|
+-----------+----------+---------+-------------+

+-----------+----------+---------+
|employee_id|department|   rating|
+-----------+----------+---------+
|        101|        IT|Excellent|
|        102|        HR|     Poor|
|        103|   Finance|     Good|
|        104|        IT|     Poor|
|        105|     Sales|Excellent|
|        106| Marketing|  Average|
|        107|   Finance|     Poor|
|        108|        HR|     Good|
|        109|        IT|  Average|
|        110|     Sales|     Poor|
+-----------+----------+---------+
```

Step 2:

Code:

```
 15  joined_df=employees_df.join(performance_df,on="employee_id",how="inner")
 16  joined_df.show()
 17
```

Output:

```
+-----------+----------+---------+-------------+----------+---------+
|employee_id|first_name|last_name|    full_name|department|   rating|
+-----------+----------+---------+-------------+----------+---------+
|        101|     Aarav|   Sharma| Aarav Sharma|        IT|Excellent|
|        102|     Priya|    Patel|         NULL|        HR|     Poor|
|        103|     Rohan|    Mehta|  Rohan Mehta|   Finance|     Good|
|        104|    Ananya|    Verma|         NULL|        IT|     Poor|
|        105|    Vikram|    Joshi| Vikram Joshi|     Sales|Excellent|
|        106|      Neha|   Kapoor|         NULL| Marketing|  Average|
|        107|     Arjun| Malhotra|Arjun Malhotra|   Finance|     Poor|
|        108|     Sneha|     Iyer|         NULL|        HR|     Good|
|        109|     Karan|    Desai|  Karan Desai|        IT|  Average|
|        110|     Meera|     Nair|         NULL|     Sales|     Poor|
+-----------+----------+---------+-------------+----------+---------+
```

Step 3:

Code:

```
17
18  enrich_df=joined_df.withColumn("full_name",when(col("full_name").isNull(),
19              concat_ws(" ",col("first_name"),col("last_name"))).otherwise(col("full_name")))
20  enrich_df.show()
21
```

Output:

```
+-----------+----------+---------+-------------+----------+---------+
|employee_id|first_name|last_name|    full_name|department|   rating|
+-----------+----------+---------+-------------+----------+---------+
|        101|     Aarav|   Sharma| Aarav Sharma|        IT|Excellent|
|        102|     Priya|    Patel|  Priya Patel|        HR|     Poor|
|        103|     Rohan|    Mehta|  Rohan Mehta|   Finance|     Good|
|        104|    Ananya|    Verma| Ananya Verma|        IT|     Poor|
|        105|    Vikram|    Joshi| Vikram Joshi|     Sales|Excellent|
|        106|      Neha|   Kapoor|  Neha Kapoor| Marketing|  Average|
|        107|     Arjun| Malhotra|Arjun Malhotra|   Finance|     Poor|
|        108|     Sneha|     Iyer|  Sneha Iyer|        HR|     Good|
|        109|     Karan|    Desai|  Karan Desai|        IT|  Average|
|        110|     Meera|     Nair|  Meera Nair|     Sales|     Poor|
+-----------+----------+---------+-------------+----------+---------+
```

Step 4:

Code:

```
22  fil_df=enrich_df.filter(col("rating")=="Poor").select("full_name","department","rating")
23  fil_df.show()
```

Output:

```
+--------------+----------+------+
|     full_name|department|rating|
+--------------+----------+------+
|   Priya Patel|        HR|  Poor|
|  Ananya Verma|        IT|  Poor|
|Arjun Malhotra|   Finance|  Poor|
|    Meera Nair|     Sales|  Poor|
+--------------+----------+------+
```